

ORAL DEFENSE PRESENTATION & LIVE DEMO SCRIPT

Turnkey Turn-by-Turn Defense Guide for Daniela S. Ungab & Rogie P. Bacanto

Notre Dame of Midsayap College • College of Information Technology and Engineering (CITE)

Canonical Title: *Receipt or Deceit: A Cross-Architecture Analysis of Convolutional Neural Network Models in Detecting Forged Digital Transaction Receipts*

PHASE 1: OPENING & TITLE PRESENTATION (Slide 1)

ACTION: Display Slide 1 with formal title. Both speakers stand straight and address the panel formally.

SPEAKER (Daniela / Rogie):

"Good morning, members of the panel, our thesis adviser Ms. Doris Ann Mariano, and guests. We are Daniela Ungab and Rogie Bacanto, fourth-year Bachelor of Science in Computer Science students. Today, we present our thesis proposal entitled: **Receipt or Deceit: A Cross-Architecture Analysis of Convolutional Neural Network Models in Detecting Forged Digital Transaction Receipts.**"

PHASE 2: THE PROBLEM & LOCAL CONTEXT (Slides 2-3)

ACTION: Switch to problem statement slide showing digital payment statistics and GCash fraud advisory.

SPEAKER:

"According to the Bangko Sentral ng Pilipinas, digital payments accounted for over 52.8% of retail transactions in 2024. In municipalities like Midsayap, micro-merchants, local online sellers, and student entrepreneurs rely on downloadable GCash receipts to verify payments before releasing items. However, fraudsters exploit this reliance by using photo editing tools or fake receipt generators to alter amounts and reference numbers. Furthermore, when receipts are sent over messaging apps like Messenger, heavy JPEG compression masks visual tampering, making manual inspection unreliable. Our study addresses this vulnerability by developing an automated, cross-architecture CNN forensic evaluation system."

PHASE 3: RESEARCH QUESTIONS & METHODOLOGY (Slides 4-6)

ACTION: Switch to Statement of the Problem slide showing SOP 1 to 5.

SPEAKER:

"Our research specifically investigates five questions: First, what is the classification performance of Basic CNN, ResNet50, and MobileNetV2 in terms of Precision, Recall, and F1-score? Second, what is their accuracy in detecting digitally edited versus programmatically generated receipts? Third, is there a significant difference in end-to-end processing time and memory load? Fourth, how does heavy social media compression impact accuracy? And fifth, which architecture offers the most practical balance of accuracy and efficiency for merchant hardware?"

EXPLAINING TO PANEL: *If the panel asks about sampling: 'We use purposive sampling to collect authentic receipts from local GCash transactors with electronic consent, and construct paired forged samples across six attack vectors.'*

PHASE 4: LIVE FORGE GUARD SYSTEM DEMONSTRATION

ACTION 1: Open browser to <http://localhost:8501> (or forgeguard.streamlit.app). The app loads with the *Classify a Receipt* screen.

SPEAKER:

"We will now demonstrate our working prototype, ForgeGuard. On the main screen, a merchant can upload a downloadable transaction receipt or evaluate standard benchmark samples."

ACTION 2: Click the 'Authentic Receipt' sample button. The system processes in ~22 milliseconds.

SPEAKER:

"When an authentic receipt is evaluated, the Error Level Analysis engine re-encodes the image at Q=90 JPEG quality. Notice under the ELA Residual Matrix tab that the compression noise is completely uniform across the canvas. As a result, Basic CNN and MobileNetV2 immediately verify the receipt as Authentic with high confidence."

ACTION 3: Click the 'Edited Tampering' sample button. Point to the Tamper Heatmap tab.

SPEAKER:

"Now we load a digitally edited receipt where the transaction amount was altered. Notice the immediate difference: under the ELA Matrix and Tamper Heatmap Overlay, the modified amount region glows with high-frequency compression variance spikes because it underwent double JPEG compression. The models evaluate the 128x128 ELA tensor and classify it as Forged within milliseconds."

ACTION 4: Scroll down to point to the 'Tri-Spectral Forensic Evidence Decomposition' gallery and the 'How the AI Analyzes This Receipt' telemetry panel.

SPEAKER:

"Beneath the results, our Tri-Spectral Decomposition displays the original document, the raw ELA noise matrix, and the thermal hotspot overlay side-by-side. The telemetry bar provides real-time mathematical signals—Mean Residual Energy, Spatial Variance, and Peak Error—demonstrating the exact physical parameters driving the neural classification."

ACTION 5: In the sidebar, select 'Model Comparison'.

SPEAKER:

"Finally, navigating to our Model Benchmark Suite reveals the empirical cross-architecture comparison across both Standard and Compressed conditions. Our findings indicate that the lightweight Basic CNN achieves 92.92% standard and 99.56% compressed accuracy at just 22.51 ms, while ResNet50 over-fits on residual noise, classifying all inputs as forged. This proves our hypothesis that compact architectures are superior for residual compression forensics."

PHASE 5: CONCLUSION & PANEL Q&A DEFENSE TIPS

ACTION: Conclude presentation formally. Both members step forward.

SPEAKER:

"In conclusion, ForgeGuard demonstrates that lightweight CNNs paired with Error Level Analysis provide an effective, real-time defense against mobile receipt fraud without requiring expensive GPU hardware. Thank you, and we are now ready to address the questions of the esteemed panel."

KEY DEFENSE RESPONSE FORMULAS:

- **If asked about CCJE students:** 'Sir/Ma'am, for validating the technical system and statistical instruments, we use CITE faculty. However, CCJE students can serve as human document examiners to rate visual realism using our rubric. If criminology examiners cannot visually detect the fakes, it proves an AI tool is needed.'
- **If asked why screenshots are excluded:** 'Screenshots include phone operating system UI like battery and carrier indicators, which introduce device-dependent compression signatures. Section 1.4 delimits our study to official downloadable receipts to guarantee controlled, platform-specific forensic evaluation.'